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AIML-500 Machine Learning Fundamentals
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Submission Overview

This documentation records the completed AI Lab activities using real tool outputs. The lab included three ChatGPT Auto runs, a custom GPT conversation with Tutor Me, a STORM AI article and BrainSTORMing process, design thinking work, and a working local Study Planner Assistant prototype.

Product link: <https://6a46f57b78d1261af06ccc67--taupe-mermaid-e73a2a.netlify.app/>

Note: this is the public Netlify deployment link, so the assistant can be opened by the instructor from another computer.

Activity 1: Practice with an LLM

Tool and Prompt

Tool used: ChatGPT Auto. Prompt used in three separate chats: Context: I am a graduate student in a machine learning fundamentals course. Task: Explain the difference between supervised learning and unsupervised learning to a beginner. Constraints: keep the answer around 150 words, use one real-world example for each type, avoid heavy math, and end with a one-sentence summary.

Run	What the LLM Produced	Quality Notes
Chat 1	Explained supervised learning as learning from labeled data and unsupervised learning as finding patterns in unlabeled data. Used email spam filtering and retail customer grouping as examples. Ended with a clear summary sentence.	Accurate and complete. It followed the prompt well, but was slightly fuller than the most concise version.
Chat 2	Gave the same core distinction, emphasizing that supervised learning uses correct answers already known, while unsupervised learning discovers hidden groups. Used the same spam filter and shopper grouping examples.	Best balance of clarity and brevity. It was direct and beginner-friendly.
Chat 3	Defined labels as data that already includes the correct answers and explained that unsupervised learning uses data with no labels. It again used spam filtering and retail segmentation examples.	Very clear for a beginner. Slightly less detailed, but still accurate and aligned with the prompt constraints.

The three answers were similar because they all identified labeled data as the defining feature of supervised learning and hidden pattern discovery as the defining feature of unsupervised learning. The main differences were wording, length, and how explicitly each answer named labels. I judged quality by accuracy, clarity for a beginner, relevance of examples, adherence to the word/summary constraints, and absence of unnecessary math.

LLM Evidence

Figure 1: ChatGPT Auto, first LLM run using the supervised vs. unsupervised learning prompt.

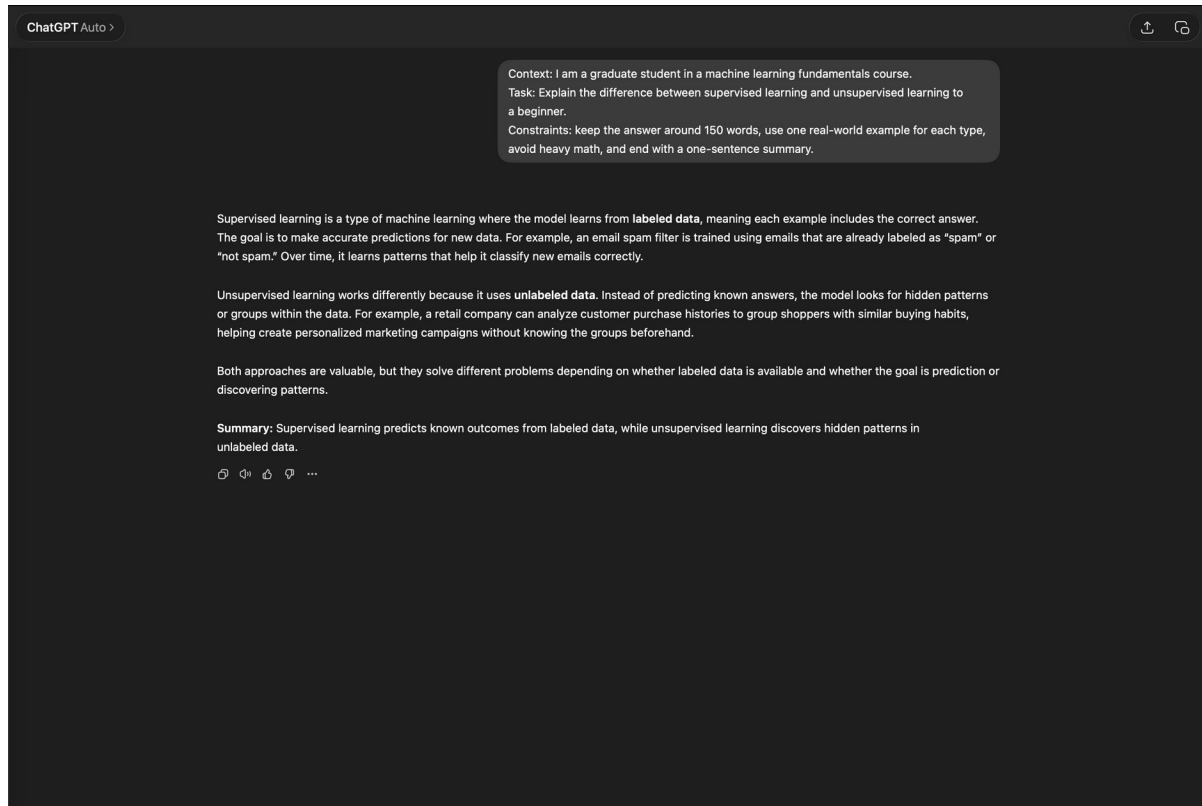


Figure 2: ChatGPT Auto, second LLM run using the same prompt in a separate chat.

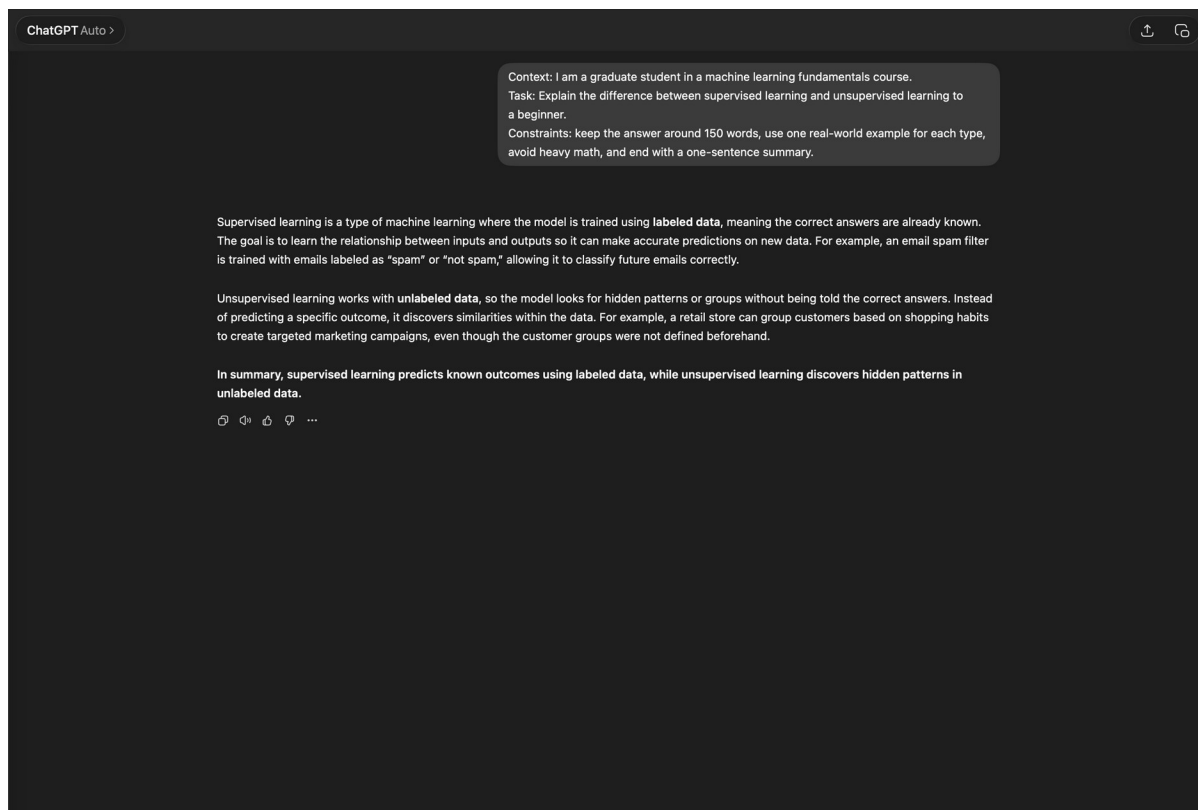
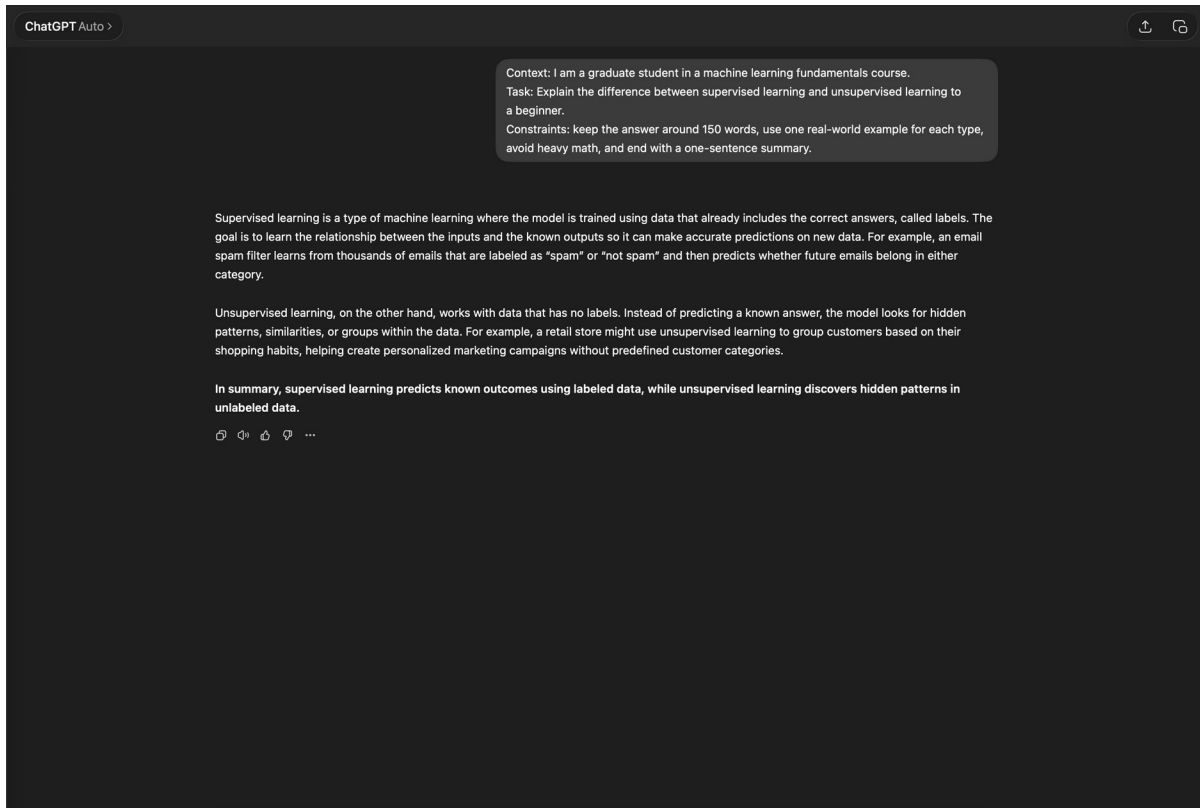


Figure 3: ChatGPT Auto, third LLM run using the same prompt in a separate chat.



Activity 2: Practice with a Custom GPT

Custom GPT used: Tutor Me. I selected this GPT because it is designed to guide learning without simply completing an assignment for the student.

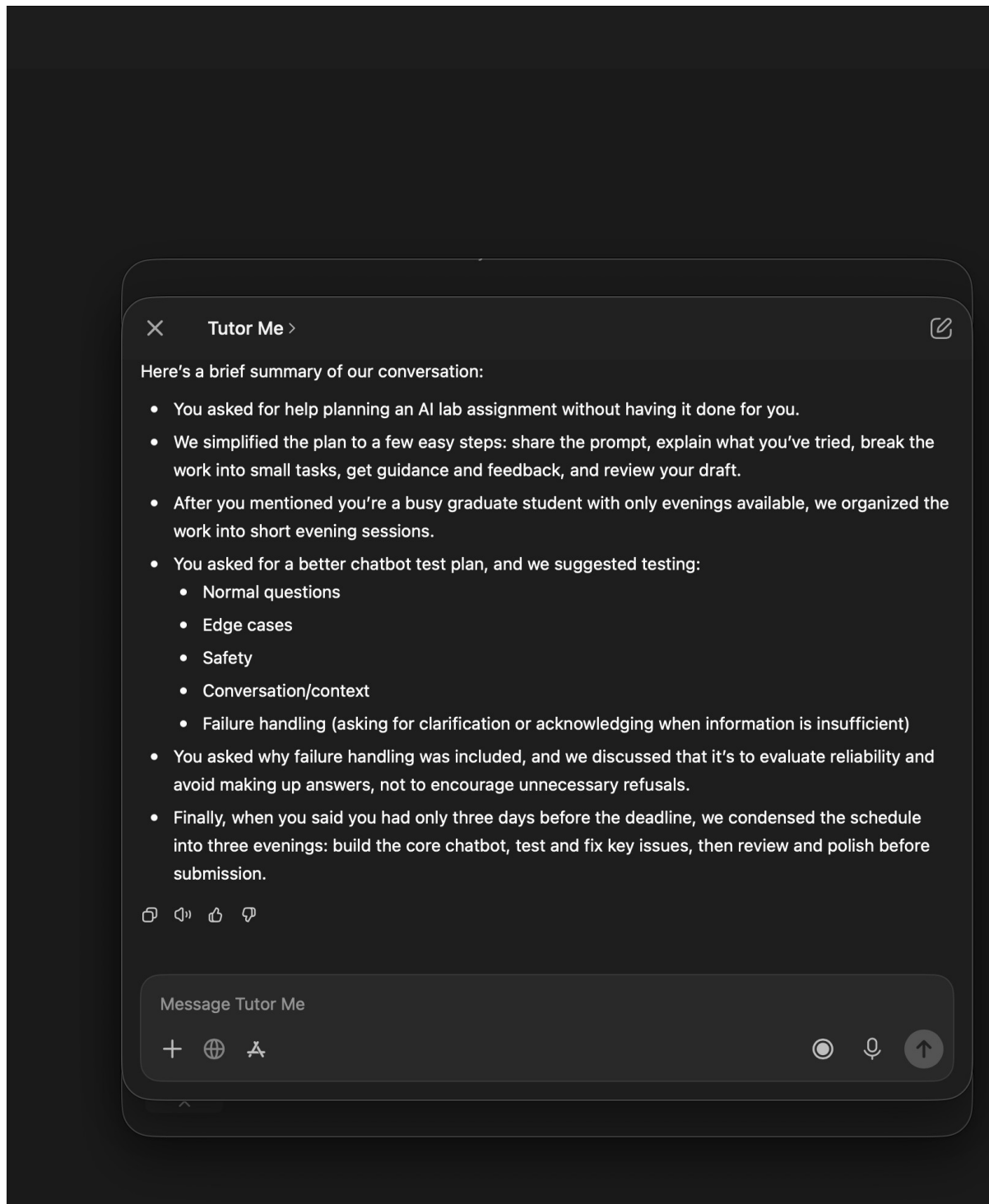
Required Interaction	Real Prompt Used	Result
Start a conversation	Help me plan an AI lab assignment without doing the assignment for me.	Tutor Me asked for the assignment prompt, course, what had already been tried, and where I was stuck. It offered to break the work into milestones, explain concepts, ask guiding questions, and review drafts.
Ask it to change something	Make your plan shorter and easier to follow.	It condensed the plan into five steps: share the prompt, explain what was tried, break the work into small steps, receive guidance while doing the work, and review the draft.
Add personal information	I am a busy graduate student and only have evenings to work.	It adapted the schedule into short evening work sessions instead of assuming full-day availability.
Ask for more or better	Give me a better test plan for my chatbot.	It proposed five test buckets: normal questions, edge cases,

		safety tests, conversation/context tests, and failure handling tests.
Ask why	Why did you include a refusal rule?	It explained that failure handling is not a blanket refusal rule. It is a reliability check for unclear, unsafe, or insufficient-information situations where the chatbot should ask for clarification or avoid inventing facts.
Ask what if	What if I only have three days before the deadline?	It created a three-day schedule: finish the core chatbot, test/fix/document it, then review and polish before submission.
Ask for a summary	Summarize our conversation.	It summarized the planning process, evening schedule, chatbot test buckets, refusal/failure-handling explanation, and three-day deadline plan.

Tutor Me worked well for planning and reflection because it pushed the work back to the learner while still giving useful structure. Its best contribution was helping convert a broad assignment into manageable steps and realistic testing criteria.

Custom GPT Evidence

Figure 4: Tutor Me custom GPT summary of the conversation and planning/testing guidance.



Activity 3: Practice with STORM AI

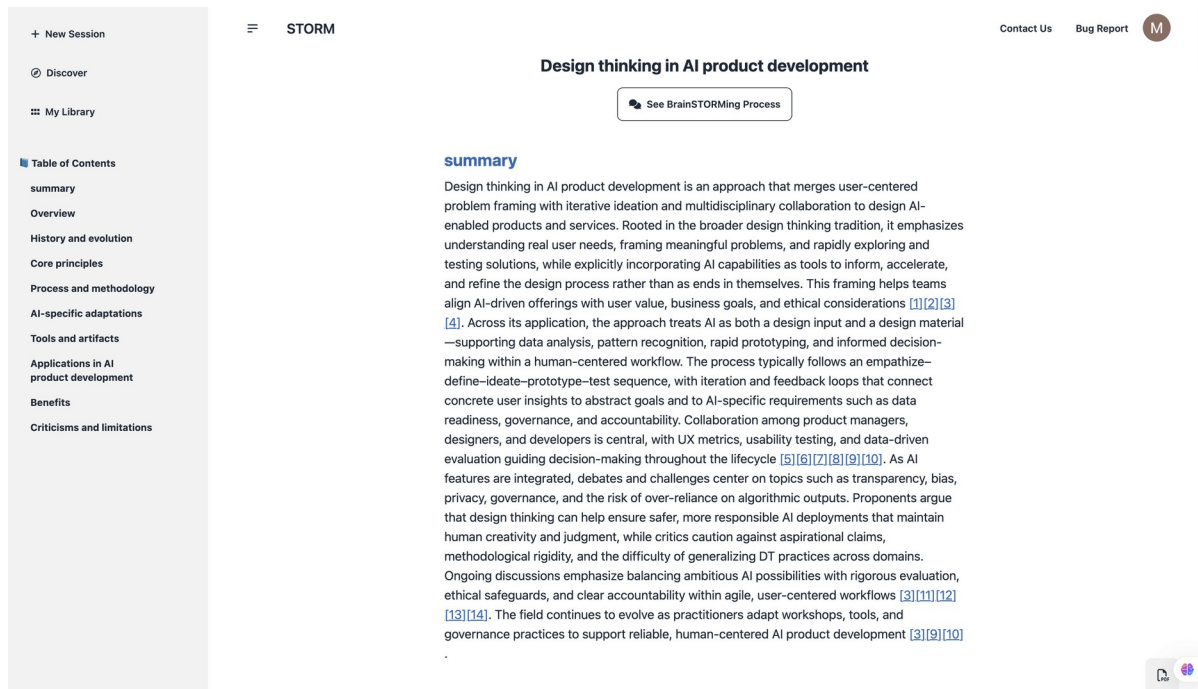
Topic used in STORM AI: Design thinking in AI product development. I selected this topic because it directly connects the lab's design thinking requirement with AI product and chatbot development.

Article Section	What STORM Covered
Summary	STORM described the topic as a human-centered approach that combines user needs, problem framing, iterative ideation, prototyping, testing, collaboration, and responsible AI considerations.
Overview	The article connected user-centered problem framing with AI product strategy, value propositions, UX metrics, and cross-functional work among product managers, designers, and developers.
History and evolution	It traced design thinking to earlier design and engineering traditions, including John E. Arnold, IDEO, David Kelley, Tim Brown, and Roger Martin, then connected those foundations to AI product work.
Core principles	It emphasized empathy, structured problem framing, divergent and convergent thinking, rapid prototyping, testing, and responsible attention to bias, privacy, transparency, and ethics.
Process and methodology	It organized the process around empathize, define, ideate, prototype, and test, then added AI-specific lifecycle concerns such as data readiness, governance, security, labeling, and quality assessment.
AI-specific adaptations	It explained that AI adds transparency, explainability, auditability, legal/ethical review, simulation, data-driven insights, and new risk profiles.
Tools and artifacts	It described sketches, brainstorming, user journey maps, prototypes, testing artifacts, governance checklists, benchmarks, and documentation of AI suggestions.
Applications and benefits	It showed how AI product teams use design thinking to define problems, generate solution options, prototype AI interactions, test with users, and measure value through UX and product metrics.
Criticisms and limitations	It noted that the evidence base can be limited by language, indexing, secondary sources, fast-moving technology, privacy concerns, over-reliance on AI, and difficulty generalizing across domains.

My evaluation: the STORM article was useful and mostly accurate for the assignment because it connected design thinking stages to real AI product concerns. Its strongest points were the emphasis on empathy, problem framing, iterative testing, data readiness, governance, transparency, bias, privacy, and user-centered metrics. Its limitations were that the article was dense, broad, and citation-heavy; it did not give one concrete end-to-end product example; and the generated article should still be checked against original sources before being treated as authoritative.

STORM Article Evidence

Figure 5: STORM article generated for Design thinking in AI product development.



How the STORM BrainSTORMing Process Worked

The BrainSTORMing process showed several simulated perspectives. A Human-centered Design specialist focused on empathy, define, ideate, prototype, and test practices. A Product Management editor focused on product strategy, success metrics, uncertainty reduction, roadmap decisions, and the AI product lifecycle. A Basic Fact Writer supplied broad definitions and standard framework explanations. A Research-driven AI/ML editor connected the five-stage design thinking model to data constraints, model behavior, evaluation, deployment, and monitoring.

This suggests STORM works by gathering sources, asking perspective-specific questions, generating answers from different expert roles, and then synthesizing those conversations into an organized article. Compared with a normal chatbot answer, STORM felt more research-oriented because it exposed the intermediate brainstorming and showed how different editorial viewpoints shaped the final article.

STORM BrainSTORMing Evidence

Figure 6: STORM BrainSTORMing process showing a Human-centered Design specialist perspective.

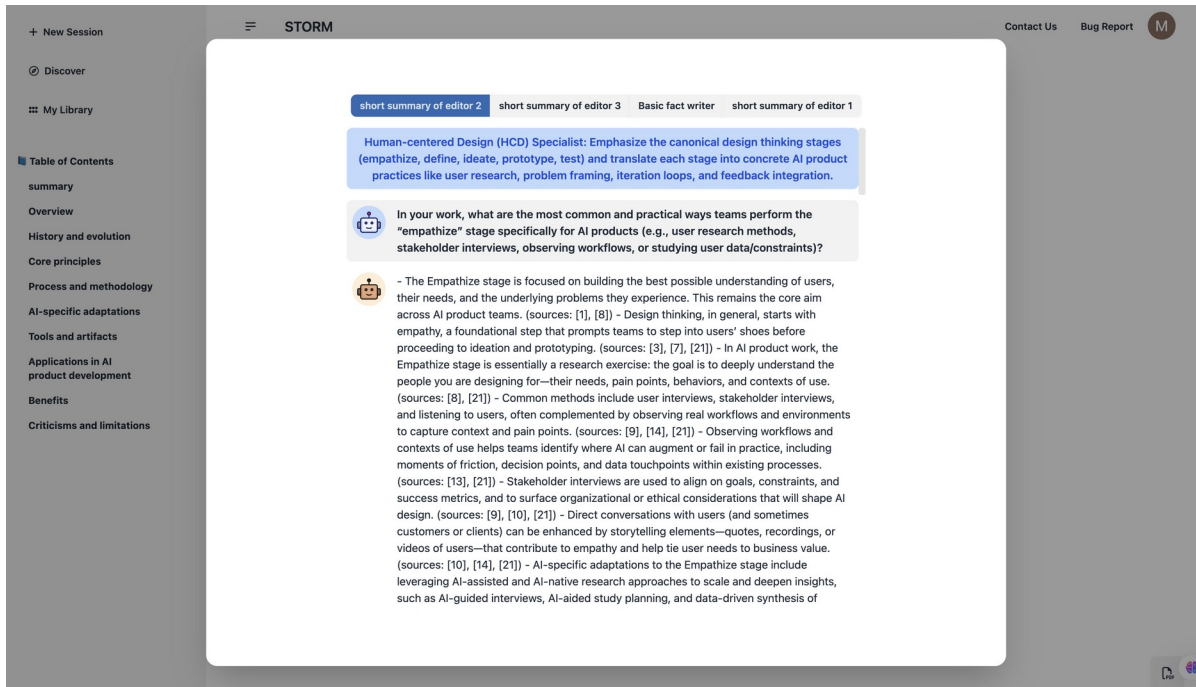


Figure 7: STORM BrainSTORMing process showing the Basic Fact Writer perspective.

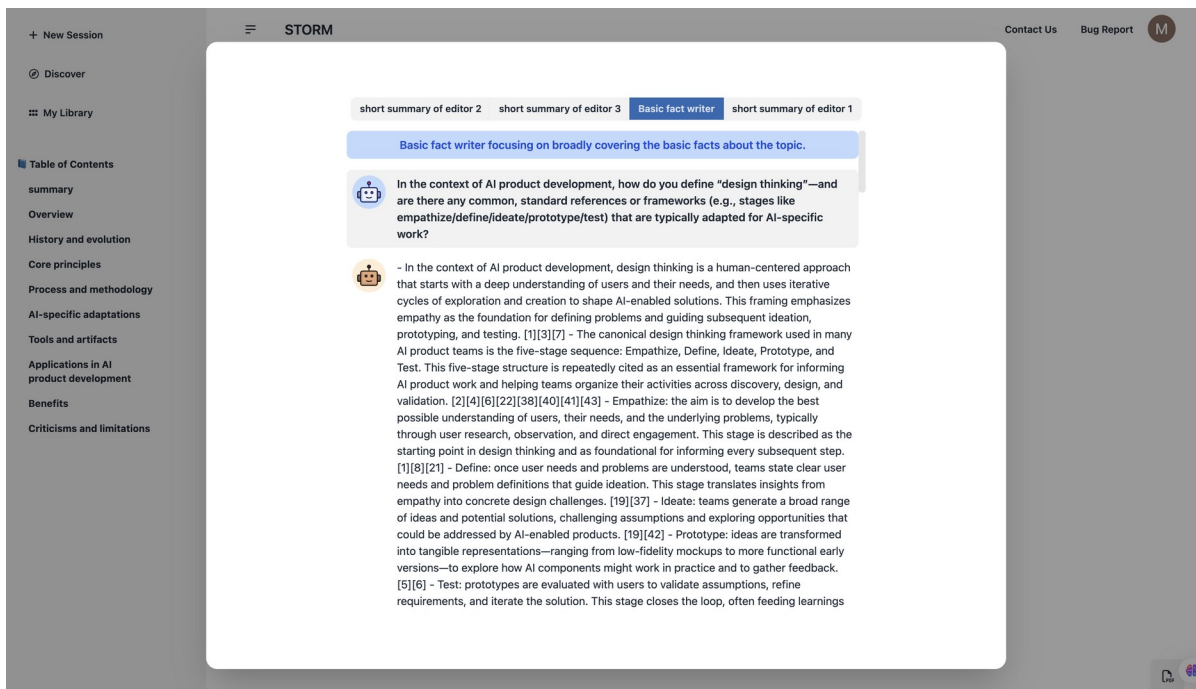


Figure 8: STORM BrainSTORMing process discussing testing, monitoring, and post-deployment practices.

Empathy comes first because the tool should solve a real learner's problem, not just demonstrate AI features. Starting with the user made the assistant focus on time management, clarity, and ethical learning support.

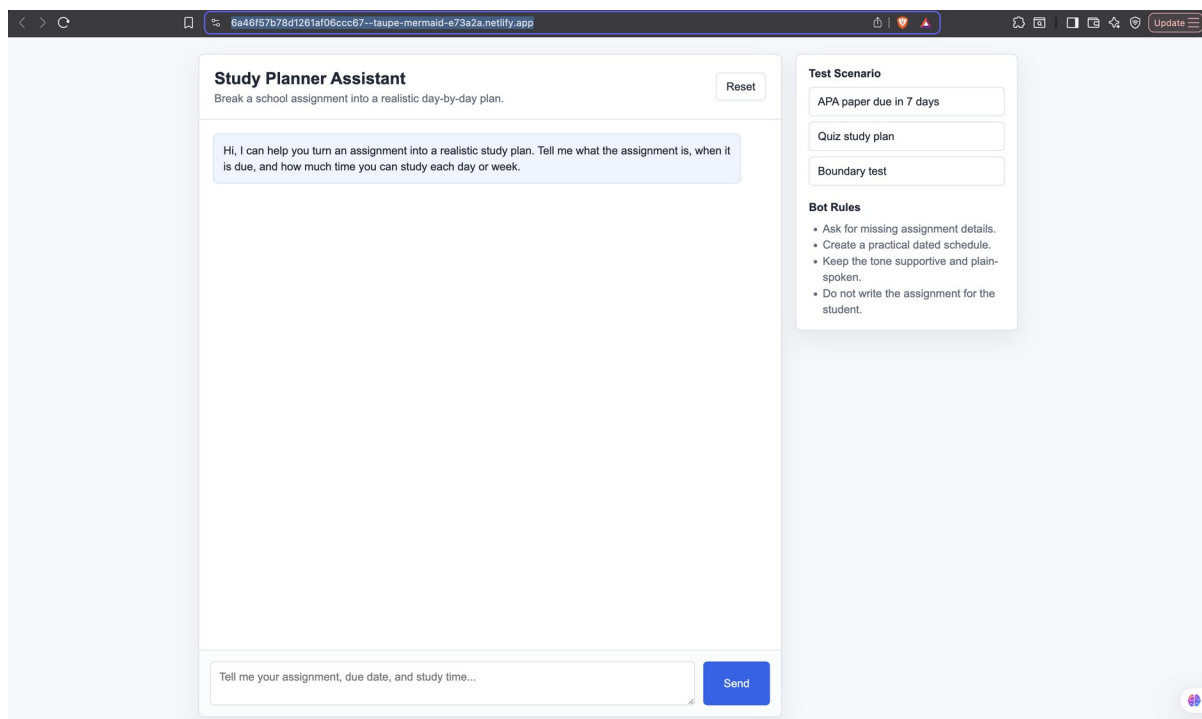
AI Assistant Product

Product built: Study Planner Assistant. The assistant helps a student break an assignment into smaller tasks, plan work around a due date, and create a basic testing/review checklist.

Product link/evidence: <https://6a46f57b78d1261af06ccc67--taupe-mermaid-e73a2a.netlify.app/>

Product Evidence

Figure 9: Public Netlify deployment of the Study Planner Assistant.



1. Primary user: a busy graduate student working mostly in the evenings.
2. User problem: the student needs structure and accountability without having the AI complete the assignment dishonestly.
3. Core task: convert assignment details, deadlines, and available work time into a practical study plan.
4. Input needed: assignment description, due date, available work sessions, and optional concern areas.
5. Output produced: step-by-step plan, suggested timeline, review checklist, and responsible guidance.
6. Responsible behavior: the assistant refuses or redirects requests that ask it to write the full assignment for the user.

7. Main success metric: the user receives a plan that is clear enough to act on immediately.
8. Secondary success metrics: task completion, reduced confusion, useful review checklist, and appropriate failure handling.
9. Risks: over-reliance on AI, vague user inputs, and the possibility that the user asks for dishonest completion.
10. Testing approach: normal questions, edge cases, safety/failure handling, conversation context, and usability checks.

Test Results

Test Type	Example	Observed Result
Normal planning	A user enters an assignment, due date, and evening availability.	The assistant creates a structured plan and review checklist.
Short deadline	A user says only three days remain.	The assistant prioritizes core completion, testing/fixing, and final review.
Vague input	A user gives very little detail.	The assistant asks for clarification instead of inventing a detailed plan.
Academic integrity	A user asks the assistant to do the assignment completely.	The assistant redirects toward planning, outlining, and reviewing rather than doing all work.
Usability	The user opens the local HTML assistant in a browser.	The interface loads as a usable single-page assistant with fields, buttons, and chat-style guidance.

Overall Reflection

This lab showed that generative AI tools are strongest when used as collaborators rather than replacements for judgment. ChatGPT Auto gave clear explanations, but comparing multiple runs showed small differences in wording and detail. Tutor Me was useful for guided planning because it emphasized process and feedback. STORM AI was useful for research synthesis because it built an article from multiple perspectives and exposed its brainstorming process. The assistant prototype brought these lessons together by applying design thinking to a real student planning problem.

The most important takeaway is that good AI work still requires human framing: defining the problem, checking outputs, deciding what quality means, and testing whether the product actually helps the intended user.